

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1773

Spatsum Lornex 138 kV
November 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1773. The Interconnection Customer proposes a 142.8 MW solar photovoltaic facility interconnecting at the Spatsum Lornex 138 kV substation in the Kentucky area. The request was received on June 07, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 521 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,855,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Kettle – Country Club 2 138 kV Line	Latah Jt Substation – Spatsum Lornex 138 kV	117.0	222	53.5
Country Club 2 – Spatsum Lornex 138 kV	Spatsum Lornex – Latah Jct 138 kV	116.0	230	53.5
Welfare – Kettle 138 kV line	Latah Jct – Spatsum Lornex 138 kV	110.1	202	38.6
Latah Jct – Alibates 138 kV	Westin-Elgin – Yuba City Sta A 138 kV	108.0	399	56.2
Spatsum Lornex – Latah Jct 138 kV	Westin-Elgin – Yuba City Sta A 138 kV	107.5	411	56.2
N Selma Substation – W Olean 138 kV	Rosendale – Noth Selma 138 kV Line	101.5	165	7.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kettle – Country Club 2 138 kV Line	Rebuild line with higher-capacity conductor	\$13,951,000	50% (\$7,019,000)
Spatsum Lornex – Country Club 2 138 kV	Rebuild line with higher-capacity conductor	\$7,323,000	50% (\$3,684,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Welfare – Kettle Substation 138 kV	Rebuild line with higher-capacity conductor	\$12,460,000	43% (\$5,388,000)
Latah Jct – Alibates 138 kV	Reconductor line	\$5,979,000	33% (\$1,993,000)
Latah Jct – Spatsum Lornex 138 kV	Reconductor line	\$1,807,000	33% (\$602,000)
North Selma – W Olen 138 kV	Reconductor line	\$3,373,000	5% (\$169,000)
	Total allocated to project		\$18,855,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.